

EX. NO: 6

IMPLEMENTATION OF DES

AIM:

To write a C program to implement Data Encryption Standard (DES) using C Language.

ALGORITHM:

STEP-1: Read the 64-bit plain text.

STEP-2: Split it into two 32-bit blocks and store it in two different arrays.

STEP-3: Perform XOR operation between these two arrays.

STEP-4: The output obtained is stored as the second 32-bit sequence and the original second 32-bit sequence forms the first part.

STEP-5: Thus the encrypted 64-bit cipher text is obtained in this way. Repeat the same process for the remaining plain text characters.

PROGRAM:

```
import java.util.Scanner;

import java.security.SecureRandom;

import javax.crypto.Cipher;

import javax.crypto.KeyGenerator;

import javax.crypto.SecretKey;

import javax.crypto.spec.SecretKeySpec;

public class Main {

    public static void main(String[] args) throws Exception {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Plain Text: ");

        String message = sc.nextLine();
```

```
KeyGenerator keyGen = KeyGenerator.getInstance("DES");
SecureRandom random = SecureRandom.getInstance("SHA1PRNG");
random.setSeed("12345".getBytes());

keyGen.init(56, random);
SecretKey secretKey = keyGen.generateKey();

byte[] raw = secretKey.getEncoded();

System.out.print("DES Key: ");
for (byte b : raw)
    System.out.printf("%02X", b);
System.out.println();

Cipher cipher = Cipher.getInstance("DES");

cipher.init(Cipher.ENCRYPT_MODE, new SecretKeySpec(raw, "DES"));
byte[] encrypted = cipher.doFinal(message.getBytes());

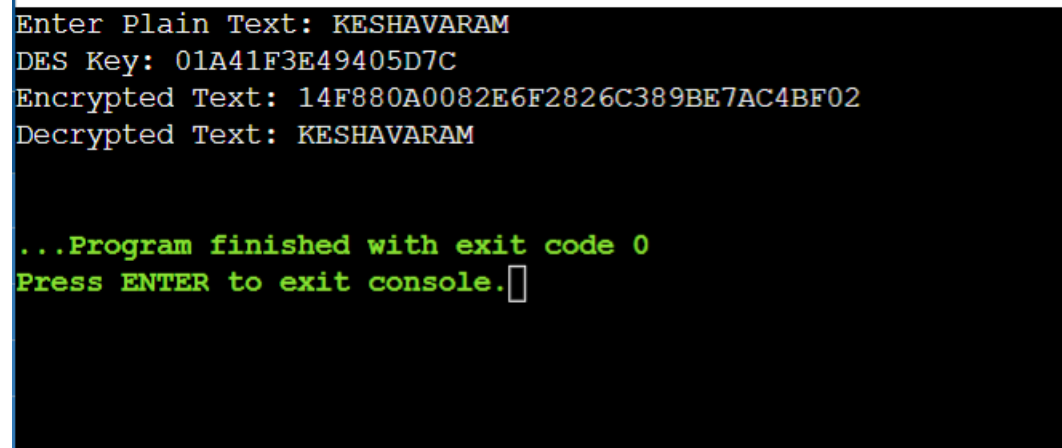
System.out.print("Encrypted Text: ");
for (byte b : encrypted)
    System.out.printf("%02X", b);
System.out.println();

cipher.init(Cipher.DECRYPT_MODE, new SecretKeySpec(raw, "DES"));
byte[] decrypted = cipher.doFinal(encrypted);

System.out.println("Decrypted Text: " + new String(decrypted));
```

```
        sc.close();  
    }  
}
```

OUTPUT:



```
Enter Plain Text: KESHAVARAM  
DES Key: 01A41F3E49405D7C  
Encrypted Text: 14F880A0082E6F2826C389BE7AC4BF02  
Decrypted Text: KESHAVARAM  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

RESULT:

Thus the data encryption standard algorithm had been implemented successfully using C language.